Abstract

1 Introduction

~~Advection – the horizontal transport of atmospheric properties – has been a subject of longstanding interest across diverse fields. Wind advectionoften transports air from one type of surface to another, for example, from a fallow area to an irrigated field. Advection from can increase evapotranspiration in fields downwind of fallowed or barren land areas, enhance~~ The surface energy balance and the associated exchange of water vapor between the land surface and the atmosphere are routinely modelled under the assumption of horizontal homogeneity, yet most land surfaces are spatially heterogeneous. Wherever surface properties change abruptly — at the boundary between an irrigated field and fallow land, at the shoreline of a lake, or at the margin of a snow-covered catchment — horizontal gradients in temperature and humidity drive the lateral transport of heat and moisture. This process, referred to here as local advection, perturbs near-surface profiles of temperature and specific humidity and, in turn, modifies the turbulent fluxes of sensible and latent heat at the surface. Its effects have been documented in a wide range of environments, including elevated evapotranspiration (ET) in irrigated fields bordered by dry land [47], enhanced evaporation from lakes and reservoirs [29], and ~~accelerate ice melting in snow-covered soils [55, 20]~~accelerated snowmelt at exposed patch margins [55, 20].

In some cases, sensible heat may be drawn from the overlying air to support evaporation, so that the latent heat flux may exceed the local net radiation (evaporative enhancement) [67, 31]. This occurs because warm, dry air moving over a cooler, well-watered surface provides an additional source of thermal energy that supplements the energy provided by net radiation [57].

While early analytical work often assumed a perfectly saturated surface [34, 47], more recent research has focused on how surface controls, such as stomatal resistance and surface relative humidity, shape energy partitioning within a soil-plant-atmosphere continuum [6].

Horizontal transport of heat and moisture across sharp surface contrasts can therefore modify near-surface gradients and, in turn, the surface turbulent fluxes.

Over heterogeneous surfaces, advection transports heat and water vapor, perturbing the one-dimensional surface energy balance assumed by most ET models.

The impact of water vapor advection is most evident near abrupt changes in land conditions, such as the edges of irrigated fields or water bodies [47, 29].

Advection and water resources

Quantifying water losses due to advection is important for water-resources planning, particularly for irrigation management in heterogeneous landscapes. Advective enhancement can be substantial, especially near leading edges where hot, dry air from adjacent fallow or barren areas increases evaporative demand over irrigated patches. Lysimeter studies show that advection can raise ET by roughly 30–10% above what local net radiation alone could support [51, 52, 57].

Consistent with this, advective effects cannot be neglected over mosaic irrigated–fallow land surfaces. This is particularly consequential during early growth stages, when seedlings are most vulnerable to high temperatures and elevated vapor pressure deficits.

Quantifying water losses due to advection is needed to effectively manage water resources. Advection is relevant to water resources planning, particularly for irrigation management. Advective enhancement is significant, especially leading edge effects. For example, lysimeter studies have shown that advection can increase ET on the order of 30–100% higher than what local net radiation alone could support [51, 52, 57]. Advective effects cannot be neglected over such variable land surfaces as mosaic irrigated/fallow lands. Quantifying advective effects is important for irrigation management. During early growth stages, seedlings are particularly susceptible to heat and high vapor pressure deficits.

Rosenberg and Verma [52] documented record-breaking evapotranspiration rates in an irrigated alfalfa field during a severe drought in the midwestern United States. On clear mid-summer days, net radiation could support at most ∼7 mm /d of ET, and sensible heat advection increased evapotranspiration to

10–14 mm/d. Allen et al (2011) observation in an irrigated agricultural area surrounded by desert, where lysimeter measurements recorded actual evapotranspiration rates that significantly exceeded the energy supplied by local net radiation. In some instances, the ratio of ET/Rn was greater than 2 meaning the actual water loss was over 100% higher than what local solar radiation alone could support. *dry air from the surrounding desert provides the additional thermal energy required to drive these extreme evaporation rates. This study demonstrates that assuming net radiation is the only source of evaporative energy can lead to significant underestimation of ET in arid and semi-arid regions.*

Land fallowing Trend towards increasing land fallowing [40]. During California drought from 2011 - 2015, for example, area of land fallowed increased from x to y?.

Previous studies examined some aspects of land use changes, driving factors, and impacts on groundwater during 2007–2016 in Central Valley. During this period, increases in fallow land area and reductions in irrigated land were reported [8,16,17].

Tools to address such questions as - how much water is saved by fallowing land in irrigated agricultural lands? Are savings to some extent offset by advection? How sensitive are things to spatial pattern?

Advection and remote sensing *Quantifying advection is also important for improving the accuracy of remote-sensing ET models. Although remote sensing advances now enable routine ET mapping, most pixel-based approaches treat each pixel as vertically one-dimensional and therefore neglect horizontal transport of heat and moisture [35, 38]. This limitation is especially problematic in advective settings, such as agricultural landscapes with patchworks of irrigated and fallow fields [17]. A small number of recent studies incorporate advection explicitly in remote-sensing ET algorithms or use high-resolution thermal imagery to diagnose advective fluxes, including modifications to SEB and SEBAL [36, 42, 65]; we discuss these in more detail in Section 5.* Quantifying advection also needed to improve accuracy of ET models. Remote sensing advances have enabled mapping ET; however, most models treat pixels as vertically one-dimensional, neglecting transport of heat and moisture [35, 38]. As a result, advective environments – such as agricultural lands with a patchwork of fallow and irrigated fields – are a challenge for ET models [17].

Remote-sensing ET models typically do not account for advection. Exceptions are several recent studies that incorporate advection into remote-sensing ET models [42, 36], including modifications to the SEB [36] and SEBAL [42]; discussed in more detail in Section 5.

Other studies have used high-resolution thermal imagery to diagnose advective fluxes [65].

Upscale in heterogeneous landscapes Advection poses an ongoing challenge in connecting high-resolution land surface data with the coarse-resolution variables used in atmospheric models [33]. Predicting how the whole differs from ths sum of its parts involves understanding how patches 'talk' to one another.

The magnitude of advective enhancement of latent heat transfer depends on the background conditions, patch configuration, and stomatal feedbacks to the advected humidity-deficit [48](Raupach, 1991; Klaassen, 1992; Kroon and de Bruin, 1993).

Advances in remote sensing This topic is particularly timely, as advances in space-based satellites, aircraft, drones, and infrared cameras have the potential to observe and analyze advective processes across various scales and conditions [64, 25, 17]. *Look for studies advancing measurement of advective effects, diagnosis using thermal imagery, drones, model/RS combination? See Kustas.* Studies using RS/aerial imagery to diagnose turbulent heat flux and advective effects...

Remote sensing technologies thus remain under-leveraged in quantifying advection and its impacts on water resources [36]. To leverage ongoing advances in remote sensing, modeling frameworks must accommodate complex, spatially variable boundary conditions derived from remote sensing, to describe advection impacts at the large scales.

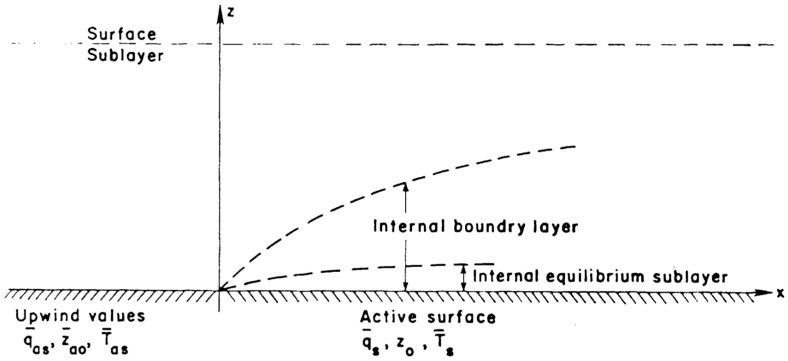

Figure 1: Sutton's problem. Definition sketch of two-dimensional internal boundary layer problem. *Placeholder*. Sutton's problem describes the idealized system of air advection across a sharp transition in surface moisture, from an upwind dry, water-limited surface to a downwind wet or irrigated surface. Upwind, strong sensible heating produces a warm, dry boundary layer. Where the surface switches abruptly to a wet, energy-limited regime. As the dry air passes over this irrigated patch, an internal boundary layer develops and grows with distance from the transition, within which temperature and humidity adjust toward values compatible with the wet surface fluxes, while the overlying flow retains the properties of the upwind dry boundary layer.

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Figure 2: Internal boundary layer of specific humidity in a dynamically uniform boundary layer.

Revisiting Sutton's problem Here, we revisit Sutton's solution [59] – describing enhanced evaporation downwind of a dry/wet interface – and outline an approach to bridge remote sensing.

Application explored is specific to mosaic fallow and cultivated areas, where highly advective conditions, experiments have demonstrated impacts.

The study first provides a brief overview of the first principles involved in the problem, including radiation balance, energy balance, mean momentum balance, and mean mass balance of water vapor.

We then examine the second principles, including the assumption of a gradient diffusion representation for turbulent fluxes. We discuss the limitation of this representation, and show that it still yields plausible results when analyzing contrasts, compared to more complex formulations [47].

Following a review of prior work / brief history.

Three sections to the study: (1) comparison to experimental results (Rider, Baldocchi) – showing good agreement. (2) estimate of fallowing effects on ET, comparing both to continuous cultivation, (3) scenario exploration.

This work advances on previous work by explicitly considering spatial configuration and wind.

2 Review of prior work

Early work

Advection is a topic rich in tradition, addressed by various fields (micrometeorology, hydrology, agronomy, boundary-layer turbulence).

Sutton's problem Sutton [59] developed an analytical solution to the humidity transport equation for the case of a sudden change in surface conditions from a dry to wet surface.

Philips (1959) Philip [44] expanded upon Sutton's 1934 work, developing an advection model describing the exchange of energy, moisture, or momentum due to horizontal heterogeneity. Philip [44]

advanced the mathematical tools to describe the advection problem by introducing a similarity variable ξ, reducing the partial differential equations of vapor transport into a ordinary differential equation, and adopting power-law profiles for mean wind speed and eddy diffusivity (allowing evaluation via computational tables). Whereas Sutton's original solution was restricted to smooth flow, Philip's approach accommodates aerodynamically rough surfaces (which are more common in natural and agricultural lands). Lastly, Sutton's solution neglected evaporation from the dry land upwind of the wet area, and Philips approach included an upwind evapotranspiration rate, which made the model more broadly applicable. The updated solutions provided a clearer way to calculate the thickness of the 'vapor blanket' (the internal boundary layer) as a function of the fetch (downwind distance). By using a power-law approximation for wind speed and eddy diffusivity, Philip provided a closed form that could be directly compared to observational results [10].

Philip-type similarity solutions remain a common starting point for footprint-style advection corrections (cite), and continue to be used in applied agricultural micrometeorology (e.g., emission/advection-correction models such as FIDES; [**?**]).

Philip's analytical approach relies on first-order closure, whereas later second-order closure models Rao et al. [e.g., 47] simulate turbulent covariances and can represent non-equilibrium adjustment, stability effects, and roughness changes across a discontinuity more mechanistically.

Energy balance constraints Studies in the 1950s–1970s extended Sutton's step-change framework by treating humidity q and temperature T as coupled through the surface energy balance, rather than as independent scalars with a fixed boundary condition. Timofeev (1954), De Vries (1959), and Rider et al. (1963) examined the simultaneous local advection of moisture and heat, and how these variables affect the atmospheric stability and surface energy partitioning. Laikhtman (1964) presented an analytical solution in which surface temperature and humidity varies along the fetch under an energy constraint (from Brutsaert). Yeh and Brutsaert [69] solved the coupled advection–diffusion system using Green's functions, while tracking how energy is partitioned downwind, including soil heat storage. These models enforced local 1-dimensional energy balance of net radiation R_n with sensible and latent heat fluxes, provided a mathematical description of advective enhancement.

Townsend (1965) – self-preservation Townsend [62, 63] introduced the assumption of self-preservation to the study of flow fields disturbed by abrupt changes in boundary condition. This approach assumes that the perturbation to the flow field caused by a change in boundary conditions is self-preserving, meaning that the disturbance profile remains shape-invariant under appropriate scaling: the flow's statistical properties, when scaled by local parameters, become independent of the initial conditions and depend only on the downstream distance x as the thickness of the internal boundary layer increases [4].

Townsend's work has received support from laboratory experiments in Antonia et al. [4], who apply a self-preserving analysis to to their heated boundary-layer experiment of a temperature disturbance that grows underneath an external turbulent boundary layer. Their experimental results for the variation of thermal-layer thickness with fetch were in close agreement with the self-preserving results obtained by integrating Townsend's (1965) formulations.

1950s - 1970s: Impact of local advection on evaporation rates and plant water status

In the 1950s - 1970s, studies broadened in scope to consider controls other than micrometeorology, in particular, the role of stomates in controlling water losses [19, 41]. During this period, a series of agricultural studies documented significant advection impacts on ET utilized precision weighing lysimeters, particularly in arid and semi-arid environments [53].

1970s onwards: Higher-order closure

Sutton's (1934) approach, and other early models, had significant limitations. Specifically, they (i) generally neglected the effects of roughness change at surface transitions, (ii) ignored thermal stratification (buoyancy effects), treating moisture as a passive scalar, and (iii) relied on assumed power-law forms for wind speed and eddy diffusivity that did not adapt to non-equilibrium conditions.

Th 1970s saw the application of higher-order closure models to address the advection problem, which addressed these limitations.

Rao et al. [47] developed a second-order turbulence closure model to simulate flow from a smooth dry surface to a grassy wet surface. The model includes equations for the mean quantities, turbulent fluxes, and the viscous dissipation rate, and solves a system of coupled partial differential equations for the mean wind speed, temperature, and humidity in the internal boundary layer.

The Rao et al. (1974) model addressed limitations in Sutton's (1934) work by incorporating surface roughness and stratification (the model explicitly accounts for buoyancy and roughness changes), and predicting the mean wind, temperature, and humidity fields such that both energy and moisture balances at the surface are satisfied. Rao et al. [47] 'showed that higher-order closure methods can become a powerful tool in the numerical analysis of evaporation problems with local advection (Brutsaert).

Subsequent field studies, such as Baldocchi and Rao [7] continued to use predictions from this 1974 model as the baseline for comparison against direct eddy-covariance measurements of momentum, heat, and moisture.

Donaldson (1973) and Mellor (1973) ...

1980s onward: Footprint analyses

In the late 1980s and early 1990s, footprint analyses, in which the flow field is specified but scalar fields vary, also became a major focus of research. Schmid and Oke (1988)... Schuepp et al. [54] presents one of the first widely used analytical models integrated into eddy-covariance software, offering a more rigorous approach to the 100:1 rule.

These models require the Eulerian velocity field (the wind profile) to be specified a priori to drive the model and predict the resulting particle trajectories of scalars like water vapor or heat [30].

Specific releases were used to test these theories... In particular, testing flow field done with point releases. Leclerc et al. (1988) used circular line sources to simulate tracer concentration fields inside plant canopies.

1990s onward

While research in the 1980s often relied on footprint analyses, where the flow field was specified and scalar fields varied, the 1990s marked a shift toward understanding feedback mechanisms between the atmosphere and biosphere. Soil, plant, and atmosphere was increasingly recognized as an integrated continuum, with atmospheric conditions and biological responses directly regulating one another. Models from this period explored how atmospheric humidity deficits influence stomatal conductance, and how this alters the energy available for sensible and latent heat [6].

Baldocchi and Rao [7] found that the advection of hot dry air did not enhance surface evaporation rates near the upwind edge of the potato field, because of negative feedbacks among stomatal conductance, humidity-deficits, and evapotranspiration. Thus, the advection of dry air over actively transpiring vegetation may not necessarily increase evaporation if negative feedback causes stomatal closure [45, 6].

Numerical advances

In the 1990s, the coupling of the flow field and scalar variables also gained attention.

Researchers increasingly used advances in simulation approaches – such as large-eddy simulation (LES) or direct numerical solution (DNS) of the Navier–Stokes equations – to investigate turbulent flow over surfaces with different roughness statistics, secondary circulations and energy balance closure. LES emerged as a tool for upscaling because it can dynamically simulate how surface heterogeneity impacts the atmospheric boundary layer [1].

Also in this period, [1] combined remote sensing with LES to study the interaction between heterogeneous land surfaces and the atmospheric boundary layer. Kustas and Albertson [32] combined LES and remote sensing data, allowing for for detailed representation of soil and vegetation contributions to mass and energy exchanges.

LES studies to follow up on: 2010s and 2020s (e.g., Eder et al., 2015; Kenny et al., 2017; Zhou et al., 2023).

2.1 Late 1990s onward: field experiments and advances in measurement

Measurements of turbulence during advection.

The 1990s saw advances in measuring turbulence structure and advective terms with multi-tower and multi-level systems in agricultural lands [7] , forests [58], mountains [27, 70], and coastal transitions [2, 49]. These advances were enabled by the maturation and widespread adoption of reliable fast-response 3-D sonic anemometry and gas analyzers, increasing standardization of eddy-covariance methods [26], and improved data logging and storage. In addition to these technical advances, coordinated field programs and networks provided the funding and coordination for large flux-network field campaigns (e.g., AmeriFlux, EuroFlux, FLUXNET, and related campaigns).

2.2 Late 1990s onward: Advection in other fields

During this period, advection began to receive attention beyond boundary-layer meteorology, including in studies of ice, snow, and soil–atmosphere interactions. *This period was marked by increasing attention to advection in other fields, including ice, snow, and soil interactions.* For example, Sutton's solution was adapted by Higgins et al. [29] to examine water vapor advection near a lake–land transition using lidar data, showing good agreement between Sutton's solution and data.

Advection of sensible heat from bare soil to snow patches. Patch-scale boundary-layer models in Shook and Gray [56] and Essery et al. [21] show that advective sensible heat can boost melt over patch edges.

Harder et al. [28]

The problem of underestimated night-time CO_2 fluxes in forest ecosystems led to advances in measurement of advective fluxes.

Field campaigns in forests showed that nighttime CO_2 advection and storage can be similar in magnitude to the eddy flux, and can substantially alter annual NEE if included [5, 37, 70, 24, 22].

2.3 Remote sensing: energy balance closure and role of advection

A full accounting of remote sensing for problems listed in introductions still awaits.

Remote sensing observations are highly derived quantities. Observe at multiple scales. Remote sensing ecosystem stresses.

On the remote sensing side, in parallel, the remote sensing as it pertains to this problem has experienced its own growth. Nonlocal warming observed in deforestation studies [13].

2.4 Advection in remote-sensing ET models

Modifications to RS-based ET models such at SEBAL and SEB have relied on (a) calibration to tower data and assumptions about closure, (b) parametizations of warming potential of the atmosphere.

These advances have demonstrated improvement in prediction; however, these models do not consider spatial configuration. In fragmented or patchy landscapes, under arid conditions, patches talk to each other.

Remains to bridge Sutton to remote sensing.

Horizontal advection contributes to surface-energy-balance non-closure: the full control-volume budget includes storage and advective flux-divergence terms, so a residual in $R_n - G - (H + LE)$ can reflect transport in addition to measurement errors.

Table 1: Variables and constants

Symbol	Definition	Equation/Value	Units
	Variables		
q	Absolute humidity		g m^{-3}
RH	Relative humidity		g m^{-3}
T	Temperature		C
LE	Latent heat flux	$L_v\overline{w'q'}$	W/m^2
H	Sensible heat flux	$\rho_{air}c_p\overline{w'T'}$	W/m^2
R_n	Net radiation		W/m^2
G	Ground heat flux		W/m^2
$\bar{u}, \bar{v}, \bar{w}$	Wind		m/s
z_{om}	Momentum roughness length		m
z_{max}	Blending height / top of domain		m
	Constants		
L_v	Latent heat of vaporization	$2.5 \cdot 10^6$	J/kg
ρ_{air}	Density of air	1.225	kg/m^3
c_p	Specific heat of air at constant pressure	1005	J/(kg K)
k	von Karman constant	0.4	

3 Methods

The analysis/simple model follows approach outlined in Brutsaert [10], and implemented by Higgins et al. [29]. We first describe for the simple case of a single dry patch upwind of a wet/irrigated patch, then how to generalize/adapt to mosaic/2-dimensional lands. In this case, input to model are equilibrium boundary conditions far-afield either size of the transition.

3.1 Conservation equations

$q(z_{max}) = q_a$ and $T(z_{max} = T_a$ are constant at the blending height, T_{max}.

Conservation of Water Vapor

The equation for the conservation of water vapor mass describes the transport of humidity in the atmosphere [10]:

$$\frac{\partial \bar{q}}{\partial t} + \bar{u}\,\frac{\partial \bar{q}}{\partial x} + \bar{v}\,\frac{\partial \bar{q}}{\partial y} + \bar{w}\,\frac{\partial \bar{q}}{\partial z} = -\,\frac{\partial\,\overline{u'q'}}{\partial x} - \frac{\partial\,\overline{v'q'}}{\partial y} - \frac{\partial\,\overline{w'q'}}{\partial z} + D_m \nabla^2 \bar{q}. \tag{1}$$

where t is time; x, y, and z are the longitudinal, lateral, and vertical coordinates; an overbar denotes time averaging and primes denote fluctuations; $\bar{q}$ is [g m^{-3}] the mean vapor concentration; $\bar{u}$, $\bar{v}$, and $\bar{w}$ [m s^{-1}] are the mean longitudinal, lateral, and vertical velocity components; $\overline{u'q'}$, $\overline{v'q'}$, and $\overline{w'q'}$ [g m^{-2} s^{-1}] are the corresponding turbulent fluxes of water vapor; and D_m [m^2 s^{-1}] is the molecular diffusion coefficient for water vapor.

Simplifications

To simplify the advection problem across a wet–dry interface, the flow is assumed stationary ($\partial\bar{q}/\partial t = 0$), aligned with the mean wind ($\bar{v} = 0$), and laterally homogeneous ($\partial(\cdot)/\partial y = 0$), with negligible large-scale subsidence ($\bar{w} \approx 0$).

Table 2: Variables, parameters and constants

Symbol	Definition	Equation/Value	Units
Variables			
q	Absolute humidity	$q = \dfrac{e}{R_v\, T}$	kg m^{-3}
RH	Relative humidity	$RH = 100\, \dfrac{e}{e_{\mathrm{sat}}(T)}$	%
T	Air temperature	$T[\mathrm{K}] = T[^\circ\mathrm{C}] + 273.15$	K
$\overline{w'q'}$	Turbulent vertical flux of water vapor (mass)	$\overline{w'q'} = -K_q\, \dfrac{\partial \bar{q}}{\partial z}$	kg m^{-2} s^{-1}
$\overline{w'T'}$	Turbulent vertical heat flux (kinematic)	$\overline{w'T'} = -K_t\, \dfrac{\partial \bar{T}}{\partial z}$	K m s^{-1}
LE	Latent heat flux	$LE = L_v\, \overline{w'q'}$	W m^{-2}
H	Sensible heat flux	$H = \rho_{\mathrm{air}} c_p\, \overline{w'T'}$	W m^{-2}
R_n	Net radiation	$R_n = (1-\alpha)SW_{\downarrow} + LW_{\downarrow} - \epsilon\sigma T_s^4$	W m^{-2}
G_s	Ground heat flux (positive downward)	$G_s = \lambda_s(\theta)\, \dfrac{T_s - T_b}{z_b}$	W m^{-2}
$\bar{u}, \bar{v}, \bar{w}$	Mean velocity components	$\bar{u}(z) = \dfrac{u_*}{\kappa} \ln\left(\dfrac{z-d}{z_{om}}\right)$	m s^{-1}
γ	Psychrometric constant	$\gamma = \dfrac{c_p\, p}{\varepsilon\, L_v}$	Pa K^{-1}
z_{om}	Momentum roughness length	—	m
$z_{\max}$	Blending height / top of domain	—	m
Constants			
L_v	Latent heat of vaporization	2.5×10^6	J kg^{-1}
ρ_{air}	Density of air	1.225	kg m^{-3}
c_p	Specific heat of air at constant pressure	1005	J kg^{-1} K^{-1}
κ	von Kármán constant	0.4	—
σ	Stefan–Boltzmann constant	5.67×10^{-8}	W m^{-2} K^{-4}
R_v	Gas constant for water vapor	461.5	J kg^{-1} K^{-1}

We further assume high Reynolds number conditions, such that turbulent transport is much larger than molecular diffusion, $\left|\overline{u_i'q'}\right| \gg |D_m\, \partial\bar{q}/\partial x_i|$. Under a boundary-layer approximation, vertical turbulent transport dominates over horizontal turbulent transport, $\left|\partial\overline{w'q'}/\partial z\right| \gg \left|\partial\overline{u'q'}/\partial x\right|$.

The remaining streamwise turbulent-diffusion term is small relative to mean advection, $\left|\partial\overline{u'q'}/\partial x\right| \ll |\bar{u}\, \partial\bar{q}/\partial x|$ (horizontal bulk advection that is much greater than horizontal turbulent diffusion). Finally, the turbulent Schmidt number is assumed to be $\mathcal{O}(1)$, so eddy viscosity and scalar eddy diffusivity are comparable and follow the law of the wall.

The remaining terms describe Sutton's problem:

$$\bar{u}(z)\, \frac{\partial \bar{q}}{\partial x} = -\, \frac{\partial}{\partial z}\, \overline{w'q'}. \tag{2}$$

where the vertical gradient in the water vapor flux is balanced by horizontal advection (Brutsaert 1982). This is one equation with three unknowns: $\bar{u}$, $\bar{q}$, and $\overline{w'q'}$.

Under these of assumptions, the conservation equation for mean humidity reduces to a two-dimensional advection–diffusion problem in (x, z), and the boundary conditions are a step change in surface moisture at the dry–wet interface.

Note - assumption of lateral homogeneity is important when considering a 2-dimensional domain, to stitch the 1-dimensional model solutions in 2 dimensions.

Mean wind profile

Assuming that there is no velocity adjustment at boundary, the mean wind profile follows a log–law:

$$\bar{u}(z) = \frac{u_*}{\kappa} \ln\left(\frac{z-d}{z_0}\right), \tag{3}$$

where κ is von Karman's constant, u_* the friction velocity, and d the zero-plane displacement height.

Conservation of Energy

The corresponding equation for heat is:

$$\frac{\partial \bar{T}}{\partial t} + \bar{u}\frac{\partial \bar{T}}{\partial x} + \bar{v}\frac{\partial \bar{T}}{\partial y} + \bar{w}\frac{\partial \bar{T}}{\partial z} = -\frac{\partial}{\partial x}\overline{u'T'} - \frac{\partial}{\partial y}\overline{v'T'} - \frac{\partial}{\partial z}\overline{w'T'} + D_m \nabla^2 \bar{T}. \tag{4}$$

where the radiative flux divergence is assumed negligible [10]. Applying same assumptions as for humidity, Equation 4 simplifies to:

$$\bar{u}\frac{\partial \bar{T}}{\partial x} = -\frac{\partial}{\partial z}\overline{w'T'}, \tag{5}$$

so that horizontal temperature advection is balanced by the heat flux divergence.

3.1.1 Eddy diffusivity model

The $\overline{w'q'}$ is parameterized with a first order eddy-diffusivity model:

$$\overline{w'q'} = -K_q \frac{\partial \bar{q}}{\partial z} \tag{6}$$

where $K_q(z)$ [L^2/T] is the eddy diffusivity for water vapor. Combining Equations 5 and 6 yields:

$$\bar{u}\frac{\partial \bar{q}}{\partial x} = \frac{\partial}{\partial z}\left(K_q \frac{\partial \bar{q}}{\partial z}\right) \tag{7}$$

Analogous to moisture, the turbulent heat flux is represented with a first-order eddy-diffusivity model:

$$\overline{w'T'} \;=\; -K_t(z)\,\frac{\partial \bar{T}}{\partial z}, \tag{8}$$

The $\overline{w'T'}$ is parameterized with an eddy diffusivity model:

$$\overline{w'T'} = -K_t(z)\frac{\partial \bar{T}}{\partial z} \tag{9}$$

where K_t [L^2/T] represents the eddy diffusivity for heat.

Combining Equations 5 and 9 yields:

$$\bar{u}\frac{\partial \bar{T}}{\partial x} = \frac{\partial}{\partial z}\left(K_t \frac{\partial \bar{T}}{\partial z}\right) \tag{10}$$

Assuming transport mechanisms of heat and water vapor in turbulent flow are similar (Reynolds analogy), $K_t = K_q$. Both are parameterized with a gradient diffusion model (aka mixing length):

$$K_t(z) = ku_* z \tag{11}$$

The approach is based on the assumption that the turbulent flux is linearly related to the mean concentration gradients [10].

Numerical solution

With appropriate boundary conditions, Equations 7 and 10 can be solved to describe q ane T across the wet/dry transition. Equations 7 and 10 are then solved for $\bar{T}(x, z)$ and $\bar{q}(x, z)$ downwind of the transition, subject to the equilibrium far-field temperature and humidity profiles.

Briefly, upwind and downwind (far-field conditions) are specified. We have an upwind profile, with encounters state BCs (Dirichlet) at the surface and mixing height (top of the domain, here, 50 m). We then solve for the q and T profiles as a function of distance from the transition. Further details of the numerical methods outlined in Appendix B.

Stitching of boundary conditions (intuition)

How the boundary conditions are stitched together ...

The step change in surface conditions is handled by connecting two 1-D equilibria through a 2-D adjustment zone. We first compute the vertically varying equilibrium profiles that would exist over a homogeneous upwind surface (dry/fallow) and over a homogeneous downwind surface (wet/irrigated), using the surface-layer closure together with the far-field (blending-height) constraints. The upwind equilibrium profile then provides the inflow condition at the transition,

$$\bar{q}(0, z) = q_{\text{up}}(z), \qquad \bar{T}(0, z) = T_{\text{up}}(z).$$

For $x > 0$, the lower boundary condition switches to the downwind surface state (e.g., $\bar{q}(x, z_{oq}) = q_s^{\text{wet}}$ and $\bar{T}(x, z_{oh}) = T_s^{\text{wet}}$, or equivalent flux conditions), while the upper boundary remains fixed at a blending height $z_{\max}$ where the flow is assumed insensitive to the surface discontinuity:

$$\bar{q}(x, z_{\max}) = q_a, \qquad \bar{T}(x, z_{\max}) = T_a.$$

The solution can be viewed as a downwind "stitching" of profiles: mean advection carries the upwind vertical structure downstream, while turbulent mixing diffuses the influence of the new surface condition upward. Near the ground, humidity and temperature adjust toward the downwind equilibrium; aloft, air retains memory of the upwind state. The thickness of this adjustment region (the internal boundary layer) grows with fetch x, producing a smooth transition from the upwind equilibrium profile toward the downwind equilibrium profile with distance from the edge.

Numerically, because Eqs. (7)–(10) are first order in x, the streamwise coordinate acts as a marching direction: given $\bar{q}(x_i, z)$ (and $\bar{T}(x_i, z)$), we solve an implicit vertical boundary-value problem to obtain $\bar{q}(x_{i+1}, z)$ (and $\bar{T}(x_{i+1}, z)$), enforcing the surface and top boundary conditions at each x-step.

Energy balance constraints

The moisture and heat equations are coupled through the equilibrium boundary conditions. This section presents an approach to determine the moisture and temperature profiles far from transitions, and corresponding sensible and latent heat fluxes. We then describe each term in the surface energy balance and outline how it can be constrained from reanalysis and remote-sensing data.

Far from the wet/dry transition, we assume that 1-dimensional surface energy balance equation holds:

$$R_n - G_s = LE + H \tag{12}$$

where R_n [W m^{-2}] is the net radiation, G_s [W m^{-2}] is the ground heat flux, LE [W m^{-2}] is the latent heat flux, and H [W m^{-2}] is the sensible heat flux. The net radiation at the surface is given by:

$$R_n = (1 - \alpha(\theta))SW_\downarrow + LW_\downarrow - \epsilon\sigma T_s^4 \tag{13}$$

where $SW_\downarrow$ [W m^{-2}] is the incident shortwave radiation, $LW_\downarrow$ [W m^{-2}] is the incident longwave radiation, $T_{s,K}$ [K] is the surface temperature (the subscript K is used to indicate units in Kelvin), and $\alpha(\theta)$ is the surface albedo, which may decrease with increasing soil moisture, such that the net radiation is less in fallowed land than cultivated.

The longwave radiation $LW_n = LW^\downarrow - LW^\uparrow$ has upward and downward components that can be modeled as:

$$LW^\uparrow = (1 - \epsilon_s)\epsilon_a\sigma T_a^4 + \epsilon_s\sigma T_{s,K}^4 \tag{14}$$

$$LW^\downarrow = \epsilon_a\sigma T_{a,K}^4 \tag{15}$$

where σ is the Stefan-Boltzmann constant ($\sigma = 5.67 \times 10^{-8}$ W m^{-2} K^{-4}), $T_{a,K}$ [K] is the atmospheric temperature, and ϵ_s and ϵ_a are the land surface and atmosphere emissivities, respectively.

Soil emissivity is mainly controlled by soil type and surface properties [23], where soil emissivity typically range from 0.9 to 0.98. For simplicity, $\epsilon_s = 0.9$ is used here [3].

Emissivity of the clear sky atmosphere is determined from [9]:

$$\epsilon_a(T_a) = 1.24\left(\frac{e_{sat}(T_a) \cdot RH/100}{T_a}\right)^{1/7}, \tag{16}$$

where $e_{sat}(T)$ [kPa] is the the saturation vapor pressure [Pa], computed from a Clausius–Clapeyron relationship (e.g., [43]):

$$e_{sat}(T) \;=\; 610.8\,\exp\left(\frac{17.27\,T}{T + 237.3}\right). \tag{17}$$

where T is the temperature in degrees C.

Sensible and latent heat fluxes are assumed constant a sufficient distance from the transition (e.g., > 200 m), that is, $\partial\bar{T}/\partial x = 0$ and $\partial\bar{q}/\partial x = 0$.

3.1.2 Sensible and latent heat fluxes

The sensible and latent heat fluxes are described by:

$$H = \rho c_p \frac{T_s - T_a}{r_a}, \tag{18}$$

$$LE = \alpha_{PT}\frac{\Delta}{\Delta + \gamma}(R_n - G_s). \tag{19}$$

where ρ [kg m^{-3}] is the mean air density, c_p [J kg^{-1} K^{-1}] is the specific heat of dry air at constant pressure, r_a [s m^{-1}] is the aerodynamic resistance, Δ [Pa K^{-1}] is the slope of the saturation vapor pressure curve, γ is the psychrometric constant [Pa K^{-1}], defined as [43]:

$$\gamma \;=\; \frac{c_p\,p}{\varepsilon\,L_v}, \qquad \varepsilon \equiv \frac{M_w}{M_d} \approx 0.622, \tag{20}$$

where p [Pa] is the air pressure. α_{PT} is the Priestly-Taylor coefficient, which accounts for the fact that atmospheric boundary layer is typically not in equilibrium with the surface and the evaporation typically exceeds the equilibrium rate. α_{PT} represents the additional evaporation driven by the entrainment of drier air from the free atmosphere above.

Here, α_{PT} represents enhancement above equilibrium evaporation driven by boundary-layer entrainment (a vertical, non-equilibrium effect), rather than the local effect associated with horizontal advection across a dry–wet discontinuity, which is treated explicitly by the Sutton-type advection solution.

Using subscripts f to represent conditions in the fallowed land upwind of the transition (e.g., H_f, LE_f) and c to represent conditions in the cultivated field downwind of the transition (e.g., H_c, LE_c).

The sensible and latent heat fluxes are related to the mean vertical temperature and humidity gradients through the eddy-diffusivity closure (Eqs. 6–9), with:

$$H = \rho c_p \overline{w'T'} \;=\; -\rho c_p K_t \left.\frac{\partial \overline{T}}{\partial z}\right|_{z=z_o} \tag{21}$$

$$LE = \rho L_v \overline{w'q'} \;=\; -\rho K_t L_v \left.\frac{\partial \overline{q}}{\partial z}\right|_{z=z_o} \tag{22}$$

where ρ [kg m^{-3}] is air density, c_p [J kg^{-1} K^{-1}] is the specific heat of air at constant pressure, L_v [J kg^{-1}] is the latent heat of vaporization, and $z = z_0$ is the near-surface reference height (a scalar roughness length)

Moisture profile To connect LE to surface humidity, Eq. (**??**) can be rewritten

$$\frac{\partial \bar{q}}{\partial z} \;=\; -\frac{\overline{w'q'}}{K_q(z)}. \tag{23}$$

Assuming $\overline{w'q'}$ is constant with height in the surface layer, and integrating to an arbitrary height $z \geq z_{oq}$ yields the mean moisture profile:

$$\bar{q}(z) \;=\; q_s - \overline{w'q'} \int_{z_{oq}}^{z} \frac{dz'}{K_q(z')}, \qquad z \geq z_{oq}. \tag{24}$$

Integrating from the moisture roughness height z_{oq} to a reference height $z_{\max}$:

$$\int_{z_{oq}}^{z_{\max}} \frac{\partial \bar{q}}{\partial z}\, dz \;=\; -\overline{w'q'} \int_{z_{oq}}^{z_{\max}} \frac{dz}{K_q(z)}, \tag{25}$$

Rrearranging to obtain the surface humidity:

$$q_a \;=\; q_s - \overline{w'q'} \int_{z_{oq}}^{z_{\max}} \frac{dz}{K_q(z)}. \tag{26}$$

When $K_q(z) = ku_* z$, the kinematic moisture flux / turbulent flux of water vapor is given by:

$$\overline{w'q'} = -kzu^* \frac{\partial q}{\partial z} \tag{27}$$

$$\int_{z_s}^{z_{\max}} \overline{w'q'} \partial z = \int_{z_s}^{z_{\max}} -kzu^* \partial q \tag{28}$$

$$\overline{w'q'} \ln\left(\frac{z_{\max}}{z_o}\right) = -ku^*(q_a - q_s) \tag{29}$$

$$\overline{w'q'} = -ku^*(q_a - q_s) \ln\left(\frac{z_{\max}}{z_{oq}}\right)^{-1} \tag{30}$$

$$q_s = q_a + \frac{1}{ku^*} \overline{w'q'} \ln\left(\frac{z_{\max}}{z_{oq}}\right) \tag{31}$$

which recovers the neutral log moisture profile. Here z_{oq} is the scalar roughness length for moisture, and $z_{\max}$ is the height at which the far-field moisture q_a is specified.

Temperature profile We rewrite Eq. (36) as

$$\frac{\partial \bar{T}}{\partial z} \;=\; -\frac{\overline{w'T'}}{K_t(z)}. \tag{32}$$

Integrating to an arbitrary $z \geq z_{oh}$ yields the integral-form mean temperature profile:

$$\bar{T}(z) \;=\; T_s - \overline{w'T'} \int_{z_{oh}}^{z} \frac{dz'}{K_t(z')}, \qquad z \geq z_{oh}. \tag{33}$$

where $\overline{w'T'}$ is assumed constant with height in the surface layer.

Integrating from the heat roughness height z_{oh}, where $\bar{T} = T_s$, to a reference height $z_{\max}$ where, $\bar{T} = T_a$, yields:

$$\int_{z_{oh}}^{z_{\max}} \frac{\partial \bar{T}}{\partial z}\, dz \;=\; -\overline{w'T'} \int_{z_{oh}}^{z_{\max}} \frac{dz}{K_t(z)}, \tag{34}$$

$$T_a - T_s \;=\; -\overline{w'T'} \int_{z_{oh}}^{z_{\max}} \frac{dz}{K_t(z)}. \tag{35}$$

For the case that $K_t(z) = k\, u_*\, z$,

$$\overline{w'T'} \;=\; -k u_* z \frac{\partial \bar{T}}{\partial z}. \tag{36}$$

$$\int_{z_{oh}}^{z_{\max}} \frac{\overline{w'T'}}{z}\, dz \;=\; \int_{z_{oh}}^{z_{\max}} -k\, u_* \frac{\partial \bar{T}}{\partial z}\, dz. \tag{37}$$

Integrating for $\bar{T}(z)$ gives the neutral log temperature profile:

$$\bar{T}(z) \;=\; T_s - \frac{\overline{w'T'}}{k\, u_*} \ln\!\left(\frac{z}{z_{oh}}\right), \qquad z \geq z_{oh}. \tag{38}$$

Integrating to z_{max} yields the log-profile relationship

$$\overline{w'T'} \ln\!\left(\frac{z_{\max}}{z_{oh}}\right) \;=\; -k\, u_* \,(T_a - T_s), \tag{39}$$

and

$$T_s - T_a \;=\; \frac{1}{k\, u_*}\, \overline{w'T'} \ln\!\left(\frac{z_{\max}}{z_{oh}}\right). \tag{40}$$

This provides a constant-flux upwind boundary condition for the equilibrium surface layer:

$$\overline{w'T'} \;=\; -\frac{k\, u_*}{\ln(z_{\max}/z_{oh})} \big(T_a - T_s\big), \tag{41}$$

i.e., $\overline{w'T'}$ is determined from the log-law temperature profile prescribed for the upwind profile (between z_{oh} and $z_{\max}$).

3.1.3 Ground heat flux

The ground heat flux G_s is modeled using a one-layer conduction approximation [66]:

$$G_s \;=\; \lambda_s(\theta)\,\frac{T_s - T_b}{z_b}, \tag{42}$$

where G_s is positive downward (into the soil), θ is volumetric soil moisture, $\lambda_s(\theta)$ [W m^{-1} K^{-1}] is the effective soil thermal conductivity, T_b is a background soil temperature at depth z_b. T_b and z_b are modeled as...

The thermal conductivity is parameterized with the equation $\lambda = 6\theta + 0.2$ W m^2 C^{-1} [66].

3.1.4 Clausius–Clapeyron relationship

The Clausius–Clapeyron relationship can be used to link surface temperature and humidity. Assuming fixed relative humidity at the surface provides an avenue to couple temperature and water vapor.

Surface humidity and temperature are linked by an prescribed relative humidity (i.e., RH ∼60% near the surface, given well-watered). Assuming relative humidity RH (in %), the actual vapor pressure $e(T)$ [Pa] is given by:

$$e \;=\; \frac{RH}{100}\,e_{sat}(T), \tag{43}$$

where T is the temperature in (°C), and the corresponding humidity q [kg m^{-3}] (water-vapor density) is:

$$q \;=\; \frac{e}{R_v\,(T + 273.15)}. \tag{44}$$

where $R_v = 461.5$ J kg^{-1} K^{-1} is the gas constant for water vapor and T is the temperature in Celsius.

3.2 Boundary conditions

Boundary conditions depend on the application, e.g., scenario exploration about land use management and fallowing, warming atmospheric conditions, operational with/without ET model boundary conditions, and comparison to experiments. We present several formulations on boundary conditions.

In all cases, we assume equilibrium conditions far from the boundaries. The model boundary conditions are used to solve/obtain the 1-dimensional profile far upwind and downwind of the transition; we then solve advection/diffusion equations for temperature and humidity, to stitch together the boundary conditions, the model predicts advective enhancement.

We consider three cases, different boundary conditions and objectives....

The upper boundary , in all cases, involves specifying q_a and T_a at a specified 'blending' height, z_{max}, i.e., a height at which atmosphere is not impacted by surface boundary conditions.

$z_{max} = 50$ m.

$\bar{q}(z_{max}) = q_a$ and $\bar{T}(z_{max}) = T_a$.

Surface boundary. Here z_{oh} is the scalar roughness length for heat (often taken as $z_{oh} \approx z_{om}/5$),

The surface boundary conditions are $T(z_{oh}) = T_s$ and $q(z_{oh}) = T_s$, where z_{oh} is a heat/moisture roughness height, with $z_{oh} \approx z_{om}/5$ (and the roughness height for heat and moisture is assumed the same).

The temperature boundary conditions are $T(z_{oh}) = T_s$, where z_{oh} is a heat roughness height, with $z_{oh} \approx z_{om}/5$.

Case 1: Comparison to experiments

This case included for comparison to experiments. Closest to original problem formulation (i.e., presented by Sutton and Brutsaert).

- Moisture and temperature profiles from experiments, or LE and H (for purpose of comparison to experiment data).
- Net radiation not needed. e.g., for comparison to [50] and other experiments.

Sensible and latent heat fluxes given, T_s and Q_s are solved for if H and LE are prescribed.

Case 2: Known T_s from satellite

Surface temperature from remote sensing or aerial imagery. If T_a is obtained from reanalysis, T_s from remote sensing (e.g., Landsat).

Components of net radiation: incoming solar radiation from reanalysis, albedo parameterized as a function of landcover and soil moisture.

Outgoing longwave radiation computed with Eqs. 14–15, using T_s, T_a, and the soil clear-sky atmospheric emissivities.

With surface and atmospheric temperature known, compute H from Equation 21.

In this case, we assume known surface relative humidity, in which case Clausius-Clapeyron links surface temperature and relative humidity. Determine surface humidity from Clausius-Clapeyron equation, and then determine latent heat flux from Equation 22.

This formulation assumes that relative humidity (RH) in the cultivated area is high, e.g., 60% [47].

Case 2A: Operational (using ET from an existing model, e.g., openET)

ET from openET [39], sensible heat flux H is from T_s and T_a. This is the most 'operational' of the approaches. This approach does not directly use net radiation (although can be included as a sense check). Intention here is to estimate / provide a roadmap to include advective enhancement in ET models.

A straightforward approach is to use the ET model predictions for the boundary conditions in equilibrium pixels, and use the net radiation as a sense check.

Case 3: Scenario exploration

For scenario exploration – i.e., rising atmospheric temperature and humidity – surface temperature is unknown.

LE may be determined as in Case 2 – by prescribing a fixed relative humidity and using Clausius-Clapeyron equation.

Starting point: Known $SW_\downarrow$, T_a, q_a from reanalysis (perturbing q_a and T_a in scenario exploration). *Specify atmospheric forcing at a reference height from reanalysis or prescribed scenarios: $SW_\downarrow$, T_a, and q_a.* Specify land-surface parameters and subsurface terms (from soil moisture and land cover): albedo $\alpha(\theta)$ and ground heat flux $G_s(\theta)$. Also specify soil moisture.

The surface temperature is adjusted until 1-dimensional energy balance at the surface is met.

This approach involves the following steps:

- Initialize (guess) a surface temperature T_s.
- Albedo $\alpha(\theta)$ and ground heat flux $G(\theta)$ depend on soil moisture θ and land cover (e.g., fallowed or irrigated).
- Compute $LW_\uparrow$ and $LW_\downarrow$ with Equations 14 and 15.
- Compute sensible heat flux H with with Equation 21

- Compute latent heat flux LE from Equation 21 assuming fixed relative humidity (linking q_s and T_s).

This approach solves for the surface temperature T_s by iterating until the one-dimensional surface energy balance closes, with R_n evaluated at each iteration:

- Initialize (guess) a surface temperature T_s.
- Compute longwave components $LW_\uparrow$ and $LW_\downarrow$ using Eqs. 14–15, and evaluate net radiation R_n.
- Compute the sensible heat flux H using Eq. 21.
- Compute the latent heat flux LE using the kinematic moisture-flux closure, (Eq. 22) under an assumed surface relative humidity (linking q_s and T_s via Clausius–Clapeyron).

The surface temperature T_s is then updated and the above steps are repeated until $R_n - G_s - H - LE = 0$.

T_s is then adjusted until 1D energy balance condition is met ($R_n - G - LE - H = 0$).

Alternatively, the Priestly Taylor equation, Equation 19, may be used to predict LE from the net radiation. the Priestley–Taylor formulation (Eq. 19). Advantage of the Priestley–Taylor formulation is easier to parameterize sensitivity to soil moisture (in exploring advection sensitivity to soil moisture). Is α_{PT} dependence on soil moisture be simpler to parameterize than on RH? From net radiation with Equation 19, let α_{PT} depend on soil moisture. Do we have more confidence in predicting relative humidity or α_{PT}?

Non of the cases use R_n directly from reanalysis. Net radiation from reanalysis is a scale much bigger than field scale (e.g., 9 km ERA), using net radiation for from reanalysis would involve dis-aggregating or down scaling using land cover.We would want to do this if we were integrating RS approach into a 'from the ground up' ET model. In terms of integrating into an existing model, we choose to use ET predictions and back up radiation.

Baldocchi et al. 1996

First comparison. Intrafield variability of scalar flux densities across a transition between a desert and an irrigated field.

The z_0 values over the desert and the field respectively were taken as 0.005 m and 0.05 m. The surface roughness for heat and moisture z_{oh} is assumed to be $z_o/5$.

Example illustrated how estimates of advective enhancement are height-dependent: a measurement height within the internal boundary layer samples air already adjusted toward downwind surface conditions, while observations above that layer retain stronger memory of the upwind state, altering inferred gradients and the fetch dependence of the solution.

Rider et al. (1963)

Rao et al. [47] used a higher-order turbulence closure model to simulate flow from a smooth dry surface to a grassy wet surface. The model includes equations for the mean quantities, turbulent fluxes, and the viscous dissipation rate, and solves a system of coupled partial differential equations for the mean wind speed, temperature, and humidity in the internal boundary layer.

Comparison to the experiment results of Rider et al. (1963) shows reasonable results (see Figure 2, which also includes the Rao model predictions for comparison).

Rider et al. [50] have mean air temperature and water vapor concentration profile measurements to compare against. Rao et al. [47] argues that $R_n - G_s$ will be increased by some 20% when going from dry to wet soils. Comparisons includes evaporation and Bowen ratio as a function of downwind distance (Figure 7 in Rao et al).

Modify approach on how to replace the lower boundary condition to accommodate $R_n - G_s$.

3.3 Warming and drying scenarios

Adjust T_a and q_a. Increase air temperature under warming.

If net radiation is constant, this decreases $T_s - T_a$.

For a given T_a, RH_a, $SW_\downarrow$, α_f, ..., the surface temperature T_s is adjusted so that radiation balance is met.

4 Results

4.1 Comparison to experiments

4.2 Warming scenarios

Examine sensitivity to rising temperatures by increasing T_a. In this case, T_s not prescribed by rather solver for.

5 Discussion

Implications for water resources management. Water resources – how much water is saved by fallowing land in CA central valley? Are the savings to some extent, partially offset by advection? How sensitive to spatial pattern?

Relative change – perturbation in flux because of perturbation in land cover.

5.1 Extensions

The model presented here is minimal, and can be extended, in particular, in obtaining equilibrium boundary conditions (i.e., latent and sensible heat fluxes far from the transition). Simple model can be extended – more complicated models (e.g., TSEB) could be used to solve for the boundary conditions far from the transition.

Describe how stability could be handled. What would be the consequences? Enhanced or suppressed advective enhancement?

5.2 Plant atmosphere continuum, soil moisture

How would we model plants closing their stomata? Adjust model to account for stomatal response.

Advection impacts on soil moisture?

Biological Feedback Mechanisms: Baldocchi and Rao (1995) caution that vegetation does not respond unconditionally to advected energy. They observed negative feedbacks – where plants close their stomata in response to increasing vapor pressure deficit – can minimize or offset potential advective enhancement.

The horizontal transport of heat and moisture across surface contrasts influences surface fluxes through a competing dynamic processes: advection can modify the atmosphere by adding heat to enhance latent heat flux, while simultaneously humidifying the air to suppress it. Furthermore, spatiotemporal heterogeneity in the landscape generates non-local large convective eddies that are independent of local boundary layer conditions [65]; these eddies transport heat and moisture across surface boundaries, biasing the one-dimensional column approach used in traditional eddy covariance and many remote sensing models.

5.3 Larger scale, other domains

Progress on these topics is necessary for addressing larger-scale issues such as tropical deforestation, where fire-created gaps and other disturbances can lead to nonlocal warming. Advection effects of

temperature – nonlocal warming effects deforestation – advection effects can be inferred [12, 14]. Simple, process based models to infer deforestation impacts on temperatures in surrounding undisturbed forests are lacking.

Reservoirs?

5.4 Advection in RS ET models

Discuss potential to integrate into ET model frameworks...

The SEB-A model in Liu et al. [36] expresses actual ET as $LE_{\text{act}} = LE_{\text{loc}} + LE_{\text{exo}}$, where LE_{exo} is a residual term representing the contribution from horizontal advection (i..e, the difference between ET modeled by 1-dimensional energy balance and measured by flux towers). A limitation is that LE_{exo} is inferred from eddy-covariance LE, attributing the energy residual to advection (neglecting other imbalance sources) and then spatially interpolated under a diffusion/IDW assumption, making estimates sensitive to tower coverage and interpolation choices.

Why can't we just adjust $alpha_PT$ upwards? In arid or semi-arid environments where advection is prevalent, α_PT values as high as 1.4, 1.5, or even 1.55 observed, reflecting the increased evaporative demand caused by the desert-like surroundings. (cite Diaz-Espejo et al., 2005; Li and Yu, 2007). This approach used in Sebal-A paper. Doesn't account for spatial patter, leading edge. How are we going to parameterize α_PT? More reason to use openET model output.

Mkhwanazi et al. [42] adapted SEBAL to model alfalfa under strong afternoon advection, and compares the original SEBAL (no explicit advection) to the modified "SEBAL-A", which includes an advective energy term in the daily ET sub-model. This approach does not explicitly consider how dry and irrigated patches interact (i.e., the spatial configuration of patches). Rather, the landscape is a raster of Landsat pixels in an irrigated fields embedded in dry areas. Spatial heterogeneity that enters the advection parameterization comes in indirectly through pixel-specific aerodynamic roughness, derived from NDVI: SEBAL uses an NDVI-based estimate of momentum roughness length, $z_{om} = \exp(a\,\text{NDVI} + b)$, with a, b empirically determined from sampled pixels and local vegetation height measurements. The z_{om} is then embedded in the wind function used to compute drying power, so the advective enhancement varies spatially as NDVI varies across the image. Patch "configuration" is represented implicitly by the spatial NDVI/roughness pattern in the pixel grid, not by an explicit model of patch geometry or inter-patch transport.

High-resolution UAV thermal data enables estimation of energy fluxes. Montibeller (2017) and Elarab (2016) modified existing models (SEBAL and METRIC) to estimate energy fluxes using high-resolution UAV thermal data. These studies partition surface temperature into soil and vegetation components to more accurately diagnose turbulent heat flux exchange, including advection effects.

6 Conclusion

This study outlines a simple approach to estimate advective enhancement associated with wind advection, integrate Sutton's solution and remote sensing. Leading edge advective effecting on ET with minimal data requirements. This work advances on previous work by explicitly considering spatial configuration of patches.

Formulating the boundary conditions to rely on surface temperature – combined with CC linking surface temperature and relative humidity – enables estimation of advective enhancement of ET. Use case explored. The modeling approach presented here offers the potential to address basic questions regarding water savings achieved by of fallowing land to conserve water. Advantage of using LST is easily observable from satellite data at fine spatial resolution, e.g., 100 m with Landsat, higher with aerial imagery;

Lastly, study outlines an approach by which Sutton's problem can be integrated into remote sensing ET and other models (e.g., soil moisture).

Appendix A: Review

6.1 ET model comparison

6.2 Reanalysis comparison

Study	System / region	What it contributed (ET-advection relevance)
Millar [41]	Irrigated vs dry surroundings (plant water status)	Early field evidence that local advection elevates evaporative demand and influences plant water status;
Dyer and Crawford [18]	Leading-edge microclimate over irrigated grass in dry surroundings	Leading-edge observations of microclimate modification (temperature/humidity profiles) over an irrigated patch; supports near-edge enhancement and fetch dependence.
Davenport and Hudson [15]	Sudan Gezira; irrigated cotton fields + dry fallows (17 km transect)	Field transect quantifying strong leading-edge enhancement and downwind decay of evaporative demand across large irrigated–fallow mosaics; very directly relevant to irrigation-water questions.
Lang et al. [34]	Flood-irrigated rice in a semi-arid region	Lysimeter-based ET as a function of distance from edge; empirical evidence that advection can inflate ET above radiation-limited expectations and enhancement decays with fetch.
Brakke et al. [8]	Semi-arid to sub-humid zone; sensible-heat advection partitioning	Separates local vs regional sensible heat advection components; gives a concrete diagnostic framing for "advected energy supply" that can explain ET/Rn > 1 cases without requiring local energy closure to match.

Appendix B: Numerical discretization

2-D model for advection. The basic equations are conservation of water vapor mass

$$U(z)\,\frac{\partial q}{\partial x} = -\,\frac{\partial}{\partial z}\,\overline{w'q'}$$

Gradient-diffusion model for turbulent flux

$$\overline{w'q'} = -K_t\,\frac{\partial q}{\partial z} = -(kzu_*)\,\frac{\partial q}{\partial z}$$

Log-law for U(z)

$$U(z) = \frac{u_*}{k}\,\log\!\left(\frac{z}{z_0}\right)$$

The steady 2-D advection–diffusion problem is solved in (x, z) by discretizing the x (streamwise) direction and, at each point, solving an implicit 1-D vertical problem. For a scalar (humidity q or temperature T), the the governing equation are written as:

$$\frac{\partial q}{\partial x} = B(z)\,\frac{\partial}{\partial z}\left(A(z)\,\frac{\partial q}{\partial z}\right), \qquad C(z) \equiv \frac{\partial A}{\partial z}, \tag{45}$$

Table 3: Selected foundational and applied studies of advection effects on evaporation/ET

Study	System	Approach	Key contribution
Internal boundary layer concept + step-change ("Sutton problem") solutions			
Sutton [59]	Dry → wet step change	Analytical K-theory solution	Analytical treatment of scalar adjustment after an abrupt surface moisture change; Establishes the canonical step-change framework used to quantify leading-edge ET enhancement and internal boundary-layer growth.
Coupled heat + vapor transfer with surface energy constraints (highly relevant to LST boundary conditions)			
Timofeev [60] De Vries [16]	Dry–wet transitions with energy coupling	Coupled heat–moisture / energy-budget frameworks	Combining heat and water-vapor advection; Link surface humidity and temperature through surface-energy constraints (→ LST-constrained boundary conditions).
Yeh and Brutsaert [68]	Water surface under local advection	Coupled turbulent heat + vapor transfer with energy balance	Solves simultaneous heat and vapor transfer while enforcing surface energy constraints (including radiative terms and heat storage options); links q_s to T_s, foundation for using remotely sensed T_s to constrain boundary conditions.
Field observations focused on internal boundary layers / local advection for heat and moisture			
Rider et al. [50]	Dry/smooth → irrigated/vegetated	Field profiles at multiple fetches	Data showing downwind evolution of mean temperature and humidity profiles following a surface discontinuity; often used for model evaluation of internal boundary-layer development and advective enhancement.
Millar [41],	Leading-edge microclimate over land	Field micrometeorology	Observations of near-edge modification of temperature/humidity and flux structure; supports the conceptual picture of rapid near-edge flux changes and gradual recovery toward a new equilibrium downwind.
Lang et al. [34]	Irrigated rice in semi-arid region	Field ET / micrometeorology	Documents the influence of local advection on crop evapotranspiration (oasis/edge enhancement) in an irrigation-relevant setting, motivating treatment beyond 1D energy balance.
Brakke et al. [8]	Heterogeneous landscapes (heat advection)	Diagnostic partitioning	Separates local vs regional sensible heat advection components; helpful for interpreting energy-balance perturbations and advective forcing in ET studies across patchy landscapes.
Higher-order closure / non-equilibrium turbulence modeling			
Rao et al. [47]	Dry smooth → wet rough (grassy)	Second-order closure model	Demonstrates that higher-order closure can reproduce observed T and humidity profiles and diagnose downwind evolution of energy partitioning (e.g., Bowen ratio); highlights sensitivity to near-surface boundary conditions over vegetation (e.g., scalar roughness, surface RH).
Empirical lake/reservoir evaporation scaling (benchmarking advection/fetch effects)			
Brutsaert and Yu [11]	Lakes/reservoirs/pans (size dependence)	Empirical mass-transfer scaling +	Evaporation parameterizations with explicit dependence on wind and moisture

Table 4: Selected advection studies emphasized in **?**] and their contributions relevant to advective enhancement of ET.

Study	**Scale / setting**	**Approach**	**Key contribution**
Sutton [59]	Local advection over a step change (dry→wet); finite fetch	Analytical similarity / diffusion solution ("Sutton's problem")	Canonical framework for how humidity (and by analogy temperature) adjusts downwind of an abrupt surface change; foundation for "vapor blanket" / internal boundary-layer scaling and fetch dependence. Often assumes idealized profiles and closures.
Philip [44]	Heterogeneous surfaces; scalar exchange (heat/moisture/momentum)	Theory of local advection	General mathematical apparatus for scalar transport over heterogeneous surfaces; clarifies how advection + turbulent diffusion shape downwind adjustment. Typically still idealized and needs boundary conditions/closures to apply.
Rider et al. [50] (plus Millar [41]; **?**]; **?**]; **?**]; **?**])	Field/lab transitions (dry→wet; roughness/thermal contrasts)	Observations of profiles / IBL structure	Experiments documenting internal boundary-layer structure in T and q downwind of a discontinuity; provides empirical targets for models of advective ET enhancement near edges. Limited coverage of vegetation complexity and full energy-balance closure.
?]	Lakes/reservoirs of varying size (fetch/area dependence)	Empirical mass-transfer scaling	Relates bulk mass-transfer coefficient to lake area; used to benchmark/interpret Sutton-type solutions for water surfaces with finite fetch. Most directly about open-water evaporation; translation to crop ET requires canopy boundary conditions.
?]	Very shallow pans / small water bodies	Empirical mass-transfer scaling	Extends pan-evaporation scaling (strong edge/advection sensitivity) and contrasts with lake-size scaling; highlights why "small wet patches" can evaporate disproportionately due to advected energy. Not vegetation ET, but instructive for patch-size effects.
?]; **?**]; **?**]; **?**]	Roughness-change internal boundary layers	Kármán–Pohlhausen / self-preservation (momentum)	Establishes practical ways to estimate IBL growth and adjustment after abrupt roughness changes; provides analogs and scaling logic used when extending to scalars (heat/moisture). Typically treats momentum first; scalar coupling adds uncertainty.
?]	Scalar local advection over a discontinuity	Kármán–Pohlhausen extended to scalars	Extends self-preservation / integral constraints to scalar flux profiles (e.g., water vapor), giving a tractable (though approximate) way to connect evaporation enhancement to fetch and assumed eddy diffusivity forms. Conservation satisfied in an averaged sense.
Townsend [61] (and related Townsend papers)	Turbulent boundary layer response to abrupt surface changes	Self-preserving solutions (non-integral form)	Alternative self-preserving formulation that can reproduce observed disturbance growth and profiles after surface changes; relevant to building simplified IBL parameterizations used in advection problems. Results can be implicit

where $A(z)$ is an eddy diffusivity (e.g. K_t) and $B(z) = U(z)^{-1}$, where $U(z)$ is a prescribed mean wind profile.

The right hand side may be expanded as:

$$B(z)\left[A(z)\,\partial^2 q/\partial z^2 + C(z)\,\partial q/\partial z\right]$$

Finite differences (forward in x, implicit in z). The numerical solution uses a mixed explicit/implicit finite difference scheme. With longitudinal/streamwise distance x, explicit finite difference (i is for x, j is for z):

On a grid (x_i, z_j) with spacings Δx and Δz, we use a forward difference for $\partial q/\partial x$ and a centered, implicit discretization of the vertical operator evaluated at the downwind station x_{i+1}:

$$\frac{q_{i+1,j} - q_{i,j}}{\Delta x} = B_{i,j}\left[A_{i,j}\,\frac{q_{i+1,j+1} - 2q_{i+1,j} + q_{i+1,j-1}}{\Delta z^2} + C_{i,j}\,\frac{q_{i+1,j+1} - q_{i+1,j-1}}{2\Delta z}\right] \tag{46}$$

The coefficients $A_{i,j}$, $B_{i,j}$, and $C_{i,j}$ are treated as known functions of z at x-index i (determined by the U and the stability parameterization).

Tridiagonal linear system. Rearranging (46) gives, for interior nodes $j = 1, \ldots, N_z - 2$, a tridiagonal system for the unknown profile $\mathbf{q}_{i+1}$ at x_{i+1} given known profile $\mathbf{q}_i$ at x_i:

$$a_j\, q_{i+1,j-1} + b_j\, q_{i+1,j} + c_j\, q_{i+1,j+1} = r_j, \tag{47}$$

with

$$a_j = -B_{i,j}\left(\frac{A_{i,j}}{\Delta z^2} - \frac{C_{i,j}}{2\Delta z}\right), \tag{48}$$

$$b_j = \frac{1}{\Delta x} + 2B_{i,j}\frac{A_{i,j}}{\Delta z^2}, \tag{49}$$

$$c_j = -B_{i,j}\left(\frac{A_{i,j}}{\Delta z^2} + \frac{C_{i,j}}{2\Delta z}\right), \tag{50}$$

$$r_j = \frac{q_{i,j}}{\Delta x}. \tag{51}$$

This system is solved with the Thomas algorithm [46].

Boundary and initial conditions. The computation requires an inflow profile $q(0, z)$. At each x_{i+1} we apply the following (Dirichlet) conditions at the surface and at top of the domain:

$$q(x, 0) = q_{\mathrm{s}}(x), \qquad q(x, h) = q_{\mathrm{top}}(x) \approx q_{\mathrm{a}}. \tag{52}$$

These are enforced by excluding the boundary nodes from the unknown vector and absorbing their contributions into the right-hand side (e.g. modifying r_1 and r_{N_z-2}).

Temperature T is solved with the same discretization, replacing q by T.

Ground heat flux parameterization

Following the empirical model of Xiong et al. [66], thermal conductivity λ is parameterized as a function of soil moisture, with lower and upper bounds set by dry and saturated conductivities:

$$\lambda_{\mathrm{dry}} \;=\; 0.51 - 0.6\,n, \tag{53}$$

$$\lambda_{\mathrm{sat}} \;=\; \lambda_{\mathrm{s}}^{1-n}\,\lambda_{\mathrm{w}}^{n}, \qquad \lambda_{\mathrm{s}} \;=\; \lambda_{\mathrm{q}}^{q}\,\lambda_{o}^{1-q}, \tag{54}$$

where n is porosity, q is quartz fraction, $\lambda_{\mathrm{w}} \approx 0.594$ W m^{-1} K^{-1}, $\lambda_{\mathrm{q}} \approx 7.7$ W m^{-1} K^{-1}, and λ_o is a representative mineral conductivity (taken as 2–3 W m^{-1} K^{-1} depending on q; see Xiong et al. [66]).

The resulting moisture-dependent conductivity is written as

$$\lambda_s(\theta) = \lambda_{\text{dry}} + (\lambda_{\text{sat}} - \lambda_{\text{dry}})\, S_r^R + c\,(S_r - S_r^2)\exp\big[S_r(1 - S_r)\big], \tag{55}$$

where $S_r = \theta/n$ is the soil saturation, R controls the low-moisture slope and c is a constant. Following Xiong et al. [66], $c = 1.5$ and R is texture-dependent (e.g., $R \approx 1.2$ for sand, $R \approx 1.5$ for fine-textured soils, and $R \approx 2.0$ for clay).

Stability feedbacks

Stability-dependent eddy diffusivity

To incorporate stability, define the Obukhov length L and stability variable $\zeta = (z - d_0)/L$, and modify the eddy diffusivity using a stability correction $\phi(\zeta)$:

$$K_t(z) = \frac{k\, u_*\,(z - d_0)}{\phi(\zeta)}. \tag{56}$$

Use a piecewise Businger–Dyer form for the stability correction:

$$\phi(\zeta) = \begin{cases} (1 - 16\zeta)^{-1/2}, & \zeta < 0, \\ 1 + 5.2\zeta, & \zeta > 0, \end{cases} \tag{57}$$

which reduces to the neutral limit $\phi = 1$ when $\zeta \to 0$.

Iterative upwind profile under stability

Because K_t depends on stability and stability depends on turbulent fluxes, the upwind profile is updated iteratively:

1. Initialize with a neutral/log profile, e.g.

$$T^{(0)}(z) = T_s - \frac{\overline{w'T'}_s}{ku_*} \ln\left(\frac{z}{z_{oh}}\right)$$

2. Given $T^{(n)}$, we estimate $L^{(n)}$ and compute $\phi^{(n)}(z)$, then update

$$K_t^{(n)}(z) = ku_*(z - d_0)/\phi^{(n)}(z)$$

3. Away from the discontinuity, set $\partial T/\partial x = 0$ and solve

$$K_t^{(n)}\,\frac{\partial^2 T}{\partial z^2} + \frac{\partial K_t^{(n)}}{\partial z}\,\frac{\partial T}{\partial z} = 0 \tag{58}$$

   for $T^{(n+1)}(z)$ subject to boundary conditions at $z \approx z_{oh}$ and $z = h$.

4. Iterate until $\max_z |T^{(n+1)} - T^{(n)}| < \varepsilon_T$ and $|L^{(n+1)} - L^{(n)}|/|L^{(n)}| < \varepsilon_L$.

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